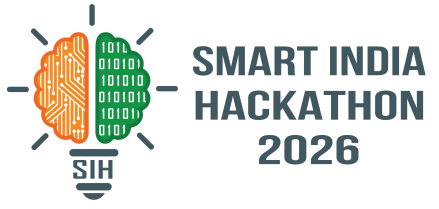
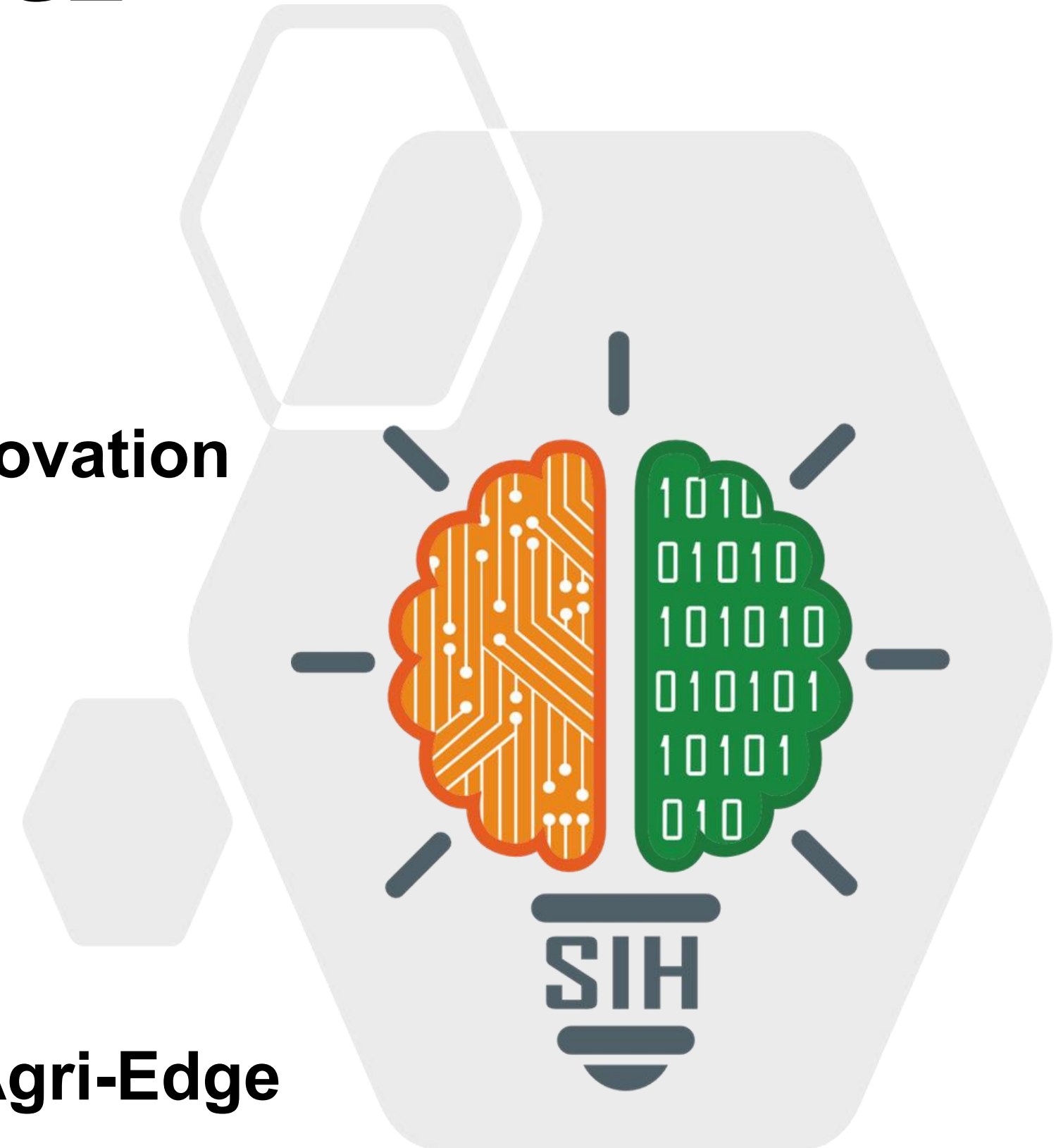


SMART INDIA HACKATHON 2026



TITLE PAGE

- **Problem Statement ID – SIH26214**
- **Problem Statement Title- Student Innovation**
- **Theme- Agriculture**
- **PS Category- Hardware**
- **Team ID- 214**
- **Team Name (Registered on portal) : Agri-Edge**



❖ Problem:

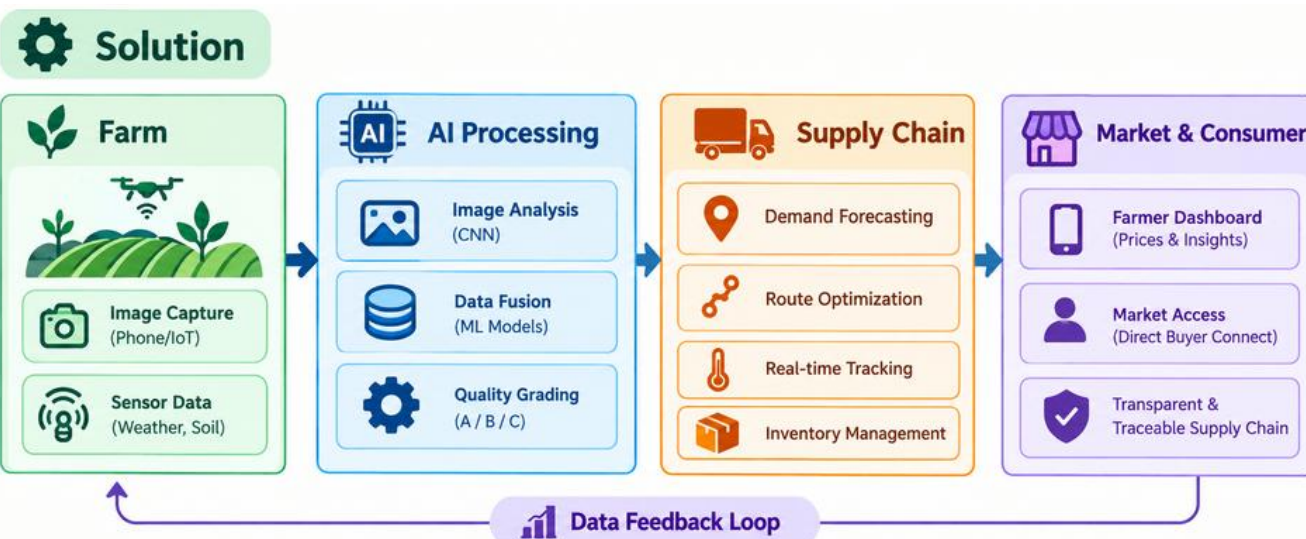
The **farmers** frequently **encounter unfair** and subjective **grading of their crops**. This results in low pricing of the crops and reliance on intermediaries. Limited access to market prices and poor tracking of the temperature and **moisture content during transportation** are also likely to **cause monetary losses** to the farmers along with wastage after harvesting.

❖ Our Idea:

The proposed solution is called **Farm to Fortune**, and it is a platform that uses artificial intelligence and Internet of Things to **grade the crops into A, B, and C categories**, provides **digital certificate** and market price comparison, and **tracks the temperature** and moisture content during **transportation**.

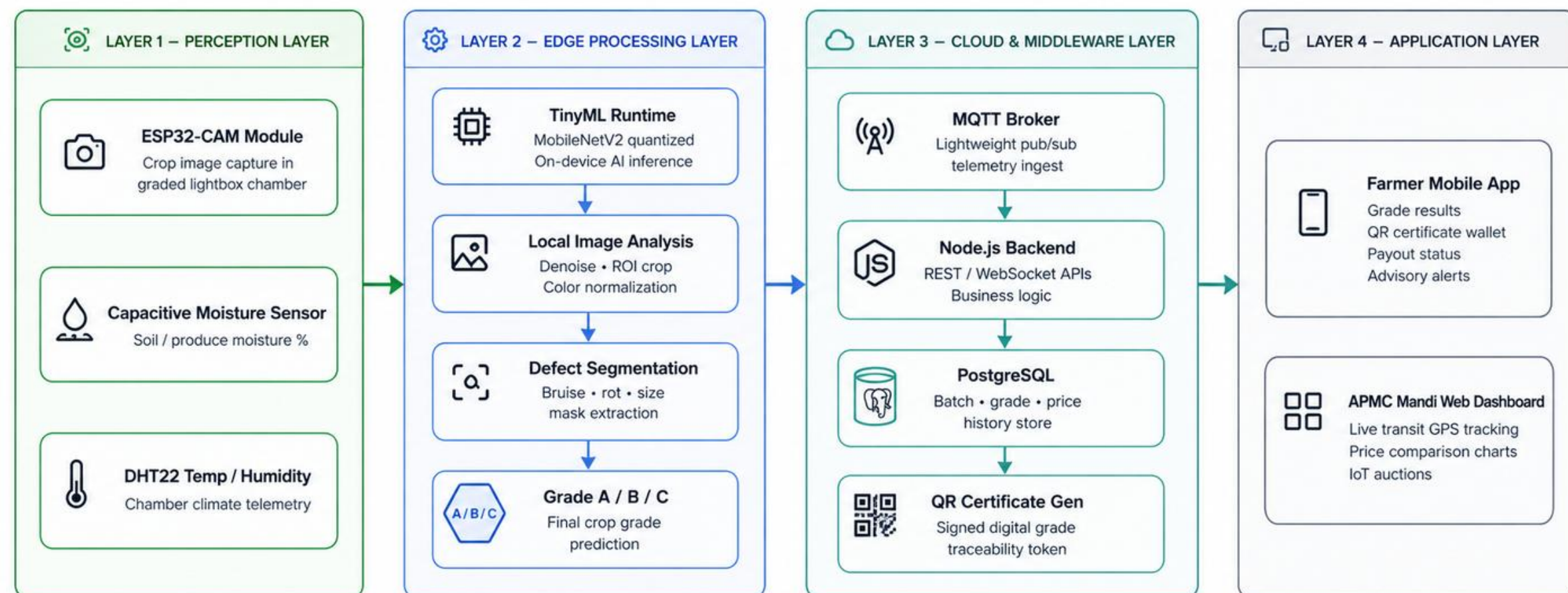
❖ Proposed Solution :

- **Objective Automated Quality Assessment:** Edge-AI powered grading stations classify crops into **A, B, and C grades** using computer vision, eliminating human bias and middlemen.
- **Reduce Post-Harvest Wastage:** Low latency monitoring tracks temperature and moisture in transit, aiming to cut wastage by 30%.



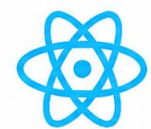
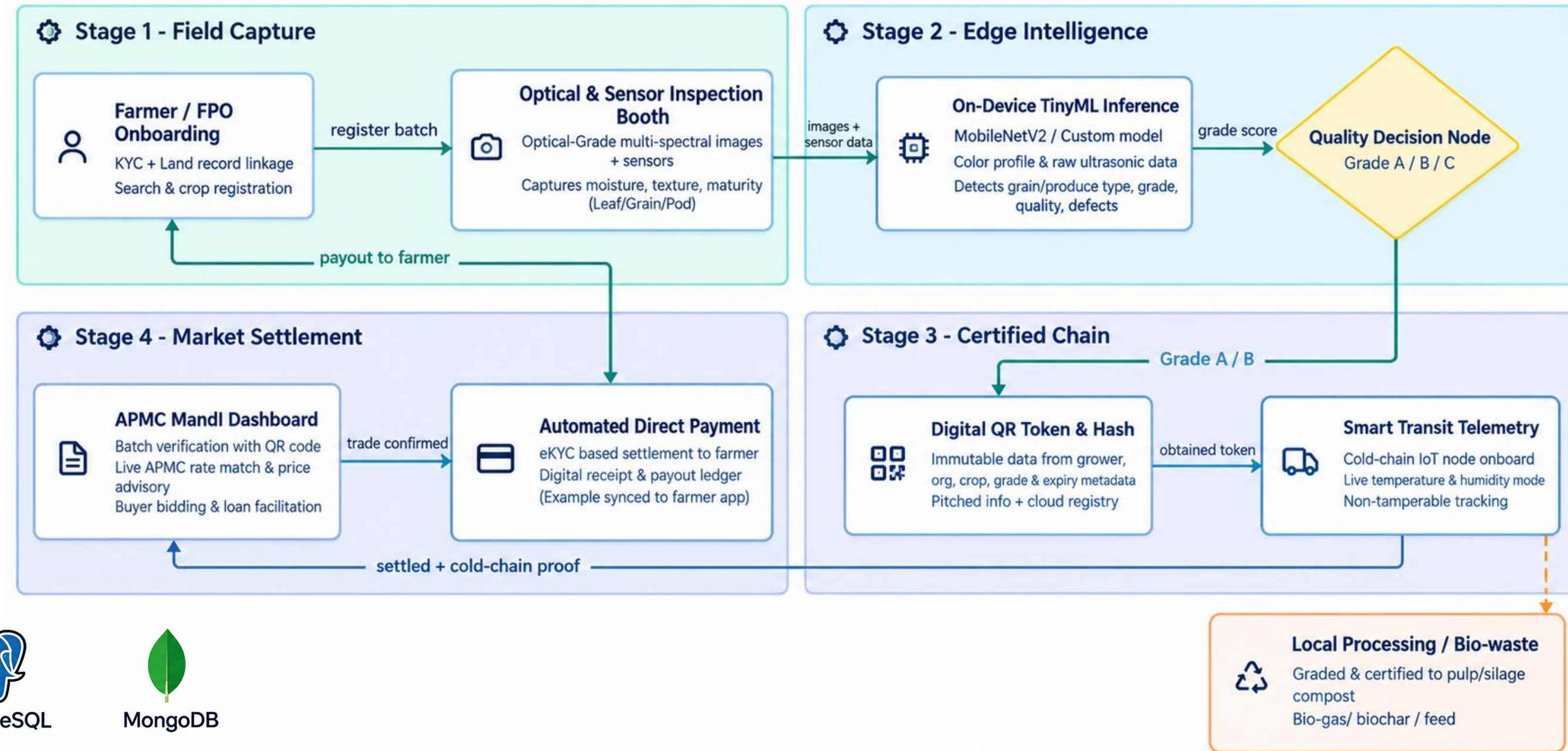
Farm-to-Fortune — System Architecture

Agri-Tech crop grading, traceability & market intelligence platform



❖ Technology Stack

- **Frontend:** React JS, React Native, Tailwind CSS.
- **Database:** PostgreSQL, MongoDB, InfluxDB.
- **Hardware:** IoT Sensors (Temperature, Humidity, Moisture), HD Cameras, ESP32 Camera, 32bit Microcontroller.
- **AI and ML:** ChatGPT 4 Model, Gemini 3.5 Turbo Model, PyTorch, OpenCV, Scikit-learn.
- **Cloud and Deployment:** GitHub, Render, Docker, Kubernetes.



React Js



React Native



Tailwind CSS



PostgreSQL



MongoDB



ESP32



32-bit Microcontroller



Python



OpenCV



TensorFlow



PyTorch



Scikit-learn



Docker



Kubernetes



AWS



Render

❖ Feasibility and Viability:

- **IoT Readiness:** Sensors make on-farm deployment practical.
- **Cloud Scalability:** AWS, Azure, GCP easily handle image and sensor data.
- **AI Model Accessibility:** Open-source frameworks (PyTorch, OpenCV, Scikit-learn) enable rapid prototyping and local inference.
- **Low Infrastructure Cost:** Uses existing connectivity reducing expenses.
- **Government & NGO Alignment:** Fits into digital agriculture and smart- farming initiatives for funding support.

❖ Challenges:

- **Connectivity Gaps:** Rural areas may face unstable internet affecting real-time uploads.
- **Data Quality Issues:** Variability in lighting & crop conditions reduce grading accuracy.
- **Farmer Adoption:** Resistance to new technology due to lack of digital literacy.
- **Regulatory Compliance:** Data privacy and blockchain standards must be met.

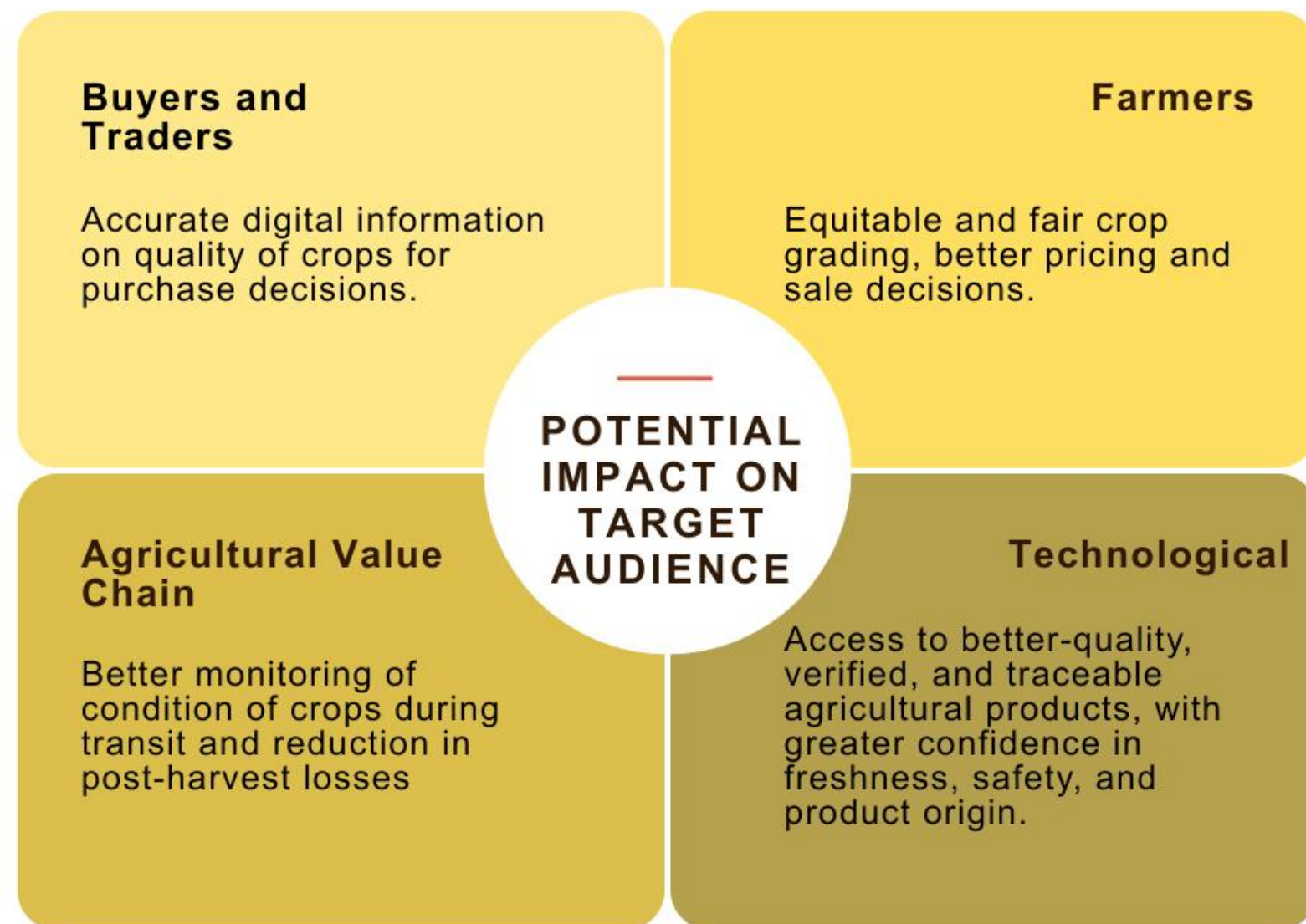
Business Potential:

- **Subscription Model:** Monthly/annual plans for farmers, buyers & logistics operators
- **Hardware + SaaS:** One-time Edge-AI/IoT setup + recurring platform fee
- **Transaction Revenue:** Small fee on verified crop sales through the platform
- **Government & Institutional Scale:** Deployment through FPOs, cooperatives, NGOs & agri-programs

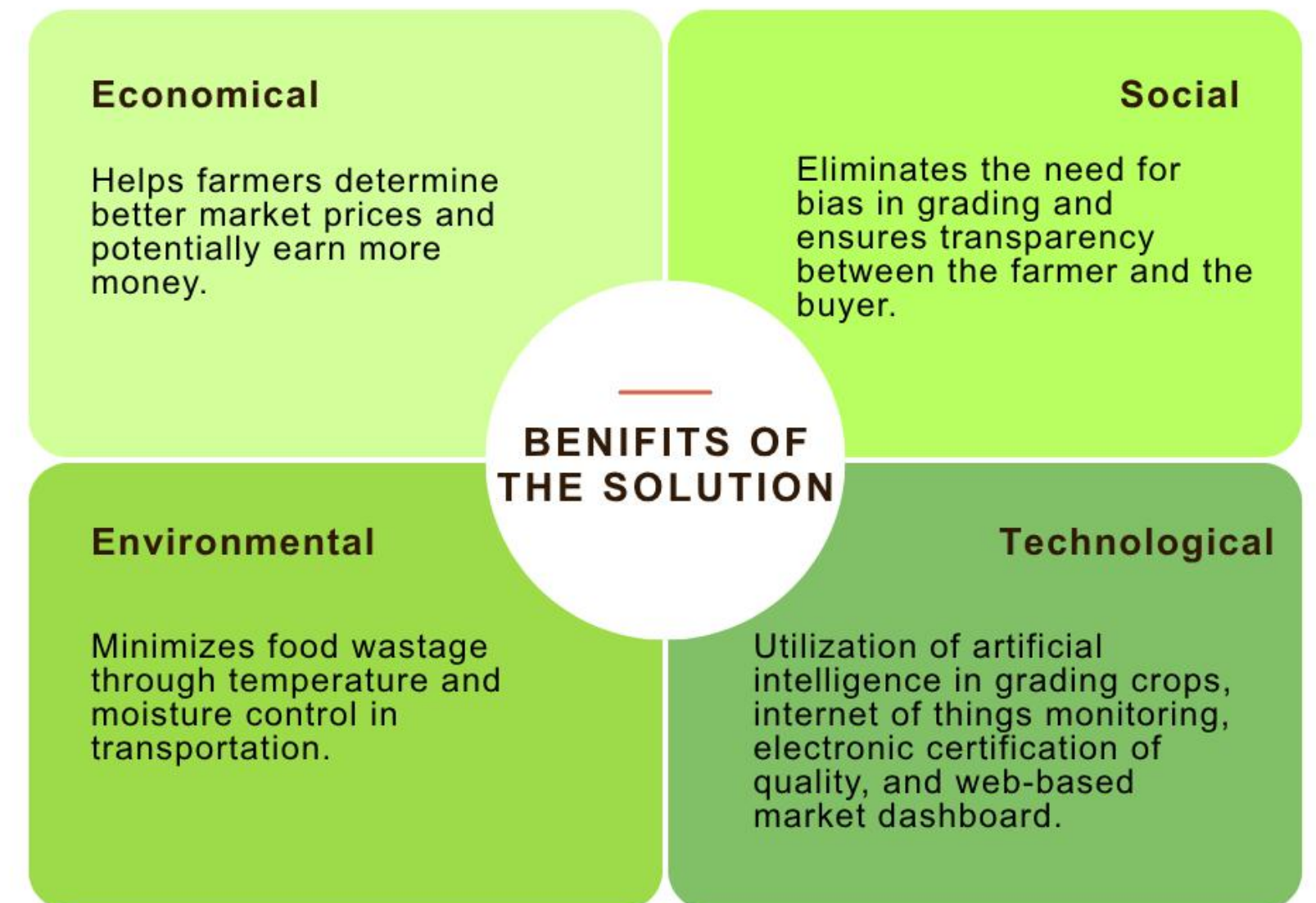
❖ Solutions:

- **Offline Data Buffering:** Store data locally and sync when connectivity resumes.
- **Adaptive AI Models:** Train models with diverse crop datasets to improve grading robustness.
- **Training & Outreach:** Conduct workshops and app tutorials for farmers.
- **Secure Data Framework:** Use encrypted cloud storage for compliance and trust.

❖ Potential Impact on Target



❖ Benefits of the Solution



❖ Crop Grading Platforms:

Agrograde : <https://www.agrograde.com/>

AgNext : <https://agnext.com/>

e-NAM : <https://enam.gov.in>

❖ Research & Best Practices:

- [AI tools for Agriculture Value Chain](#)
- [Towards automatic field plant disease recognition](#)
- [Using AI for higher education](#)
- [Real Time Supply Chain Management](#)

❖ Feasibility Facts:

- [AWS IoT Core Documentation](#)
- [GSMA - AgriTech Ecosystem Connectivity](#)

GitHub Repository:

<https://agriedge-farmtofortune.github.io/AgriEdge-Framework/>

❖ Comparison with Existing Platforms:

FEATURES	FARM-TO-FORTUNE	AGROGRADE	AGNEXT	e-NAM
AI-based Crop Grading	✓	✓	✓	×
Automatic Grade A/B/C	✓	✓	✓	×
Digital Quality Certificate	✓	×	✓	×
Market Price Comparison	✓	×	×	✓
Farmer-Buyer Connection	✓	Limited	Limited	✓
Temperature & Moisture Monitoring	✓	×	✓	×
Transportation Monitoring	✓	×	Limited	×
End-to-End Farm-to-Market Solution	✓	×	Partial	Partial
Main Technology	AI + IoT + Web Platform	AI / Computer Vision	AI + IoT + Data Analytics	Digital Marketplace Platform

Live Demo:

<https://agriedge-farmtofortune.github.io/AgriEdge-Framework/>